

INTERACTIVE 3D MODEL · SPECIFICATION

V8 5.0

Naturally aspirated racing engine

A 60-degree V8 built for circuit racing: high revs, dry-sump lubrication, an individual throttle for every cylinder and a titanium exhaust.

Type	petrol, naturally aspirated, V8, 60° bank angle
Displacement	4,999 cc
Bore × stroke	92.0 × 94.0 mm
Compression ratio	12.5 : 1
Power	≈ 470 kW (640 hp) at 7,500 rpm
Torque	≈ 610 N·m at 6,000 rpm
Valvetrain	DOHC, 32 valves, chain drive
Fuel system	port injection, 8 throttle bodies
Ignition	individual coil on every plug
Lubrication	dry sump, 4-stage pump
Firing order	1-5-4-8-7-2-6-3
Weight	≈ 150 kg dry

Figures are typical for racing V8s of this class. The model is an illustrative reconstruction from a photograph; part dimensions are approximate.

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ASSEMBLY 01 · 4 items

Cylinder block

The structural backbone of the engine. The pistons run in the block and the crankshaft turns in it; the heads, front cover, sump and accessories all bolt to it. Coolant passages and oil galleries run inside.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
1.01	Cylinder block	1	Aluminium alloy, heat-treated casting	Holds the cylinders in exact alignment and carries combustion pressure and crankshaft loads. Every other assembly bolts to it.	Bank angle: 60° Bore spacing: 106 mm Bank offset: 22 mm Casting weight: ≈ 38 kg
1.02	Cylinder liners	8	Aluminium with Nikasil coating (nickel + silicon carbide)	Form the running surface of each cylinder for the piston and rings.	Bore: 92.0 mm Coating: 0.05–0.08 mm Finish: plateau honing
1.03	Main bearing caps	5	Forged steel	Together with the block bulkheads they form the crankshaft bearing housings.	Fastening: 2 vertical + 2 cross bolts Housing bore: 64 mm
1.04	Knock sensors	2	Piezo element in a steel body	Detect knock, an explosive combustion that can destroy pistons in seconds.	Principle: piezoelectric, broadband Location: one per bank

ASSEMBLY 02 · 6 items

Crank train

Turns the up-and-down motion of the pistons into crankshaft rotation. This is where combustion pressure becomes torque, which leaves through the flywheel and clutch to the gearbox.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
2.01	Crankshaft	1	Alloy steel forging, nitrided	Collects the forces from eight connecting rods and delivers torque to the flywheel and accessory drive.	Crank radius: 47 mm (94 mm stroke) Main journals: 5 × Ø60 mm Crankpins: 8 × Ø50 mm, 30° split Weight: ≈ 17 kg
2.02	Connecting rods	8	Alloy steel, H-section	Link each piston to the crankshaft and carry force both ways: on the power stroke and when the piston is stopped at each dead centre.	Centre-to-centre length: 150 mm Weight: ≈ 560 g Bearings: tri-metal, lead bronze
2.03	Pistons	8	Forged aluminium alloy, coated skirt	Take the gas pressure in the cylinder and pass it through the pin to the connecting rod.	Diameter: 92.0 mm Compression height: 31 mm Weight with pin: ≈ 380 g Mean piston speed at 7,500 rpm: 23.5 m/s
2.04	Piston rings	8 sets of 3	Nitrided steel, cast iron	Seal the gap between piston and cylinder: gas must not get down, oil must not get up.	Ring heights: 1.0 / 1.2 / 2.0 mm Set: 2 compression + oil control
2.05	Piston pins	8	Tool steel, diamond-like coating	Hinge the piston to the connecting rod small end.	Diameter: 22 mm Length: 58 mm Retention: 2 circlips

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
2.06	Balance shaft	1	Steel, eccentric weights	Cancels the vibration a 60° V8 produces.	Speed: equal to crankshaft Direction: counter-rotating Location: in the vee of the block

ASSEMBLY 03 · 13 items

Cylinder head, right bank

Closes the right bank of cylinders from above. The head holds the combustion chambers, intake and exhaust ports, 16 valves and two camshafts that open each valve at exactly the right moment.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
3.01	Head gasket	1	Multi-layer steel with elastomer coating	Seals the joint between head and block: it holds combustion pressure and keeps coolant and oil apart.	Compressed thickness: 1.2 mm Layers: 3–5
3.02	Cylinder head	1	Aluminium alloy, permanent-mould casting, heat-treated	Forms the combustion chambers and the ports through which the mixture enters and the exhaust leaves.	Valves per cylinder: 4 Included valve angle: 30° Weight: ≈ 14 kg
3.03	Intake valves	8	Titanium alloy	Open the way for fresh mixture into the cylinder on the intake stroke.	Head diameter: 37 mm Stem diameter: 6 mm Lift: 11.5 mm
3.04	Exhaust valves	8	Nickel-based heat-resistant alloy	Let the burnt gas out of the cylinder into the exhaust manifold.	Head diameter: 32 mm Lift: 11.5 mm Head temperature: up to 800–900 °C
3.05	Valve springs	16	Silicon-chrome steel, shot-peened	Close the valve and keep the valvetrain against the cam so the valve follows the lobe exactly.	Seat load: ≈ 400 N Load at full lift: ≈ 1,000 N
3.06	Tappets	16	Steel, diamond-like coating	Pass the cam force to the valve stem and take the side load.	Diameter: 31 mm Valve clearance: 0.20–0.30 mm
3.07	Intake camshaft	1	Alloy steel, induction-hardened	Opens and closes the intake valves of the bank in step with the pistons.	Opens: 30° before TDC Closes: 60° after BDC Duration: 270° crank Valve lift: 11.5 mm
3.08	Exhaust camshaft	1	Alloy steel, induction-hardened	Controls the exhaust valves of the bank.	Opens: 70° before BDC Closes: 30° after TDC Duration: 280° crank Overlap: 60°
3.09	Camshaft bearing caps	5	Aluminium alloy	Hold the camshafts in their bearings in the head.	Fastening: 4 bolts per cap Camshaft journals: Ø26 mm
3.10	Cylinder head bolts	10	High-strength chrome-molybdenum steel	Clamp the head to the block through the gasket hard enough that gas cannot escape.	Thread: M11 × 1.5 Clamp load: ≈ 80 kN per bolt

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
3.11	Spark plugs	4	Iridium electrode, ceramic insulator	Ignite the mixture in the combustion chamber at precisely the calculated moment.	Thread: M10 × 1.0 Gap: 0.6 mm Breakdown voltage: up to 35 kV
3.12	Cam cover	1	Carbon fibre, twill weave	Covers the valvetrain, keeps the oil in and the dirt out.	Weight: ≈ 0.7 kg Process: prepreg, autoclave
3.13	Ignition coils	4	Plastic housing, epoxy potting	Turn the 14 V supply into tens of kilovolts for the spark.	Output voltage: up to 40 kV Dwell time: ≈ 2–3 ms

ASSEMBLY 04 · 13 items

Cylinder head, left bank

Closes the left bank of cylinders from above. The head holds the combustion chambers, intake and exhaust ports, 16 valves and two camshafts that open each valve at exactly the right moment.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
4.01	Head gasket	1	Multi-layer steel with elastomer coating	Seals the joint between head and block: it holds combustion pressure and keeps coolant and oil apart.	Compressed thickness: 1.2 mm Layers: 3–5
4.02	Cylinder head	1	Aluminium alloy, permanent-mould casting, heat-treated	Forms the combustion chambers and the ports through which the mixture enters and the exhaust leaves.	Valves per cylinder: 4 Included valve angle: 30° Weight: ≈ 14 kg
4.03	Intake valves	8	Titanium alloy	Open the way for fresh mixture into the cylinder on the intake stroke.	Head diameter: 37 mm Stem diameter: 6 mm Lift: 11.5 mm
4.04	Exhaust valves	8	Nickel-based heat-resistant alloy	Let the burnt gas out of the cylinder into the exhaust manifold.	Head diameter: 32 mm Lift: 11.5 mm Head temperature: up to 800–900 °C
4.05	Valve springs	16	Silicon-chrome steel, shot-peened	Close the valve and keep the valvetrain against the cam so the valve follows the lobe exactly.	Seat load: ≈ 400 N Load at full lift: ≈ 1,000 N
4.06	Tappets	16	Steel, diamond-like coating	Pass the cam force to the valve stem and take the side load.	Diameter: 31 mm Valve clearance: 0.20–0.30 mm
4.07	Intake camshaft	1	Alloy steel, induction-hardened	Opens and closes the intake valves of the bank in step with the pistons.	Opens: 30° before TDC Closes: 60° after BDC Duration: 270° crank Valve lift: 11.5 mm
4.08	Exhaust camshaft	1	Alloy steel, induction-hardened	Controls the exhaust valves of the bank.	Opens: 70° before BDC Closes: 30° after TDC Duration: 280° crank Overlap: 60°
4.09	Camshaft bearing caps	5	Aluminium alloy	Hold the camshafts in their bearings in the head.	Fastening: 4 bolts per cap Camshaft journals: Ø26 mm

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
4.10	Cylinder head bolts	10	High-strength chrome-molybdenum steel	Clamp the head to the block through the gasket hard enough that gas cannot escape.	Thread: M11 × 1.5 Clamp load: ≈ 80 kN per bolt
4.11	Spark plugs	4	Iridium electrode, ceramic insulator	Ignite the mixture in the combustion chamber at precisely the calculated moment.	Thread: M10 × 1.0 Gap: 0.6 mm Breakdown voltage: up to 35 kV
4.12	Cam cover	1	Carbon fibre, twill weave	Covers the valvetrain, keeps the oil in and the dirt out.	Weight: ≈ 0.7 kg Process: prepreg, autoclave
4.13	Ignition coils	4	Plastic housing, epoxy potting	Turn the 14 V supply into tens of kilovolts for the spark.	Output voltage: up to 40 kV Dwell time: ≈ 2–3 ms

ASSEMBLY 05 · 5 items

Intake system

Feeds air into the cylinders and meters it. Each cylinder has its own trumpet and its own throttle, so the engine responds instantly to the pedal.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
5.01	Intake trumpets	8	Composite	Guide air smoothly into the intake tract and set its tuned length.	Lip diameter: ≈ 80 mm Tract length to valve: ≈ 280 mm
5.02	Throttle bodies	2 (4 butterflies each)	Aluminium alloy, anodised	Control the amount of air, and therefore the engine output.	Bore: 52 mm Butterflies: 8
5.03	Intake runners	8	Carbon fibre	Connect the throttle bodies to the ports in the cylinder head.	Section at head: Ø48 mm Weight: ≈ 80 g
5.04	Throttle linkage	1 set	Steel, aluminium	Transfers pedal movement to the throttle shafts of both banks.	Return springs: 2, redundant
5.05	Throttle position sensor	1	Plastic housing, contactless Hall sensor	Tells the control unit how far the throttles are open.	Signal: 0.5–4.5 V Resolution: ≈ 0.1°

ASSEMBLY 06 · 4 items

Fuel system

Delivers petrol at constant pressure to eight injectors, which spray it into the intake runners in exactly the amount the control unit calculated.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
6.01	Fuel rails	2	Aluminium alloy, anodised	Distribute fuel to the injectors of each bank and smooth out pressure pulses.	Working pressure: 4 bar Inner diameter: 12 mm
6.02	Fuel injectors	8	Solenoid valve in a steel body	Spray petrol into the intake runner of each cylinder.	Flow rate: ≈ 650 cc/min Pulse width: 2–14 ms
6.03	Fuel pressure regulator	1	Anodised aluminium, diaphragm	Keeps rail pressure constant and returns surplus fuel to the tank.	Setting: 4.0 bar Range: 2.5–6 bar
6.04	Fuel lines and AN fittings	1 set	Stainless-braided hoses, anodised fittings	Feed fuel to the rails, link the rails together and return surplus to the regulator.	Size: AN-6, ≈ 8 mm bore Seal: 37° cone

ASSEMBLY 07 · 5 items

Exhaust manifold, right bank

Collects exhaust gas from the four cylinders of the right bank into a single pipe. Pipe lengths and diameters are chosen so that pressure waves help scavenge the cylinders.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
7.01	Manifold flange	1	Stainless steel	Clamps the manifold pipes to the cylinder head and carries the whole assembly.	Thickness: 8 mm Fastening: 8 × M8 studs
7.02	Primary pipes	4	Titanium, 0.9 mm wall	Carry the gas from each cylinder to the collector.	Diameter: 44 mm Tuned for: 6,000–7,500 rpm Gas temperature: up to 900 °C
7.03	4-into-1 collector	1	Titanium	Merges the four primary pipes into one.	Outlet diameter: 76 mm Joint: V-band clamp
7.04	Lambda sensor	1	Zirconia ceramic in a steel body	Measures the oxygen in the exhaust, which shows the mixture strength.	Type: wideband, 5-wire Operating temperature: ≈ 750 °C
7.05	Studs and nuts	8	Heat-resistant steel, copper-plated nuts	Fasten the manifold flange to the head.	Thread: M8 × 1.25 Torque: ≈ 22 N·m

ASSEMBLY 08 · 5 items

Exhaust manifold, left bank

Collects exhaust gas from the four cylinders of the left bank into a single pipe. Pipe lengths and diameters are chosen so that pressure waves help scavenge the cylinders.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
8.01	Manifold flange	1	Stainless steel	Clamps the manifold pipes to the cylinder head and carries the whole assembly.	Thickness: 8 mm Fastening: 8 × M8 studs
8.02	Primary pipes	4	Titanium, 0.9 mm wall	Carry the gas from each cylinder to the collector.	Diameter: 44 mm Tuned for: 6,000–7,500 rpm Gas temperature: up to 900 °C
8.03	4-into-1 collector	1	Titanium	Merges the four primary pipes into one.	Outlet diameter: 76 mm Joint: V-band clamp
8.04	Lambda sensor	1	Zirconia ceramic in a steel body	Measures the oxygen in the exhaust, which shows the mixture strength.	Type: wideband, 5-wire Operating temperature: ≈ 750 °C
8.05	Studs and nuts	8	Heat-resistant steel, copper-plated nuts	Fasten the manifold flange to the head.	Thread: M8 × 1.25 Torque: ≈ 22 N·m

ASSEMBLY 09 · 7 items

Camshaft drive

Links the crankshaft to the four camshafts. Two chains turn the cams at exactly half crankshaft speed and keep the valve timing precise at any rpm.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
9.01	Front cover	1	Aluminium alloy, casting	Encloses the camshaft chain drive, keeps the oil in and carries the water pump and accessories.	Bolts: ≈ 30 Weight: ≈ 4.5 kg
9.02	Camshaft drive chains	2	Alloy steel	Transfer rotation from the crankshaft to the camshafts of each bank.	Pitch: 8 mm Ratio: 2 : 1
9.03	Camshaft sprockets	4	Steel, hardened	Take the drive from the chain and turn the camshafts.	Teeth: 36 Pitch diameter: ≈ 92 mm
9.04	Crankshaft drive sprocket	1 (double)	Steel, hardened	Drives both camshaft chains from the nose of the crankshaft.	Teeth: 2 × 18 Mounting: key and central bolt
9.05	Chain guides and tensioner shoes	4	Polyamide on a steel backing	Guide the chains and damp their side-to-side motion.	Sliding face: polyamide Per chain: 2
9.06	Hydraulic chain tensioners	2	Aluminium body, steel plunger	Keep the chains at constant tension.	Plunger travel: ≈ 12 mm Supply: engine oil pressure
9.07	Camshaft position sensors	2	Hall sensor in a plastic housing	Tell the control unit which stroke each cylinder is on.	Type: Hall effect Marks per shaft: 4

ASSEMBLY 10 · 7 items

Accessory drive

The units the engine drives itself: the alternator and the water pump, both turned by a single poly-V belt from the crankshaft pulley.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
10.01	Torsional vibration damper	1	Steel, elastomer	Damps crankshaft torsional vibration and drives the accessory belt.	Diameter: 164 mm Belt grooves: 6
10.02	Poly-V belt	1	Synthetic rubber, aramid cord	Transfers rotation from the crankshaft pulley to the alternator and water pump.	Profile: PK, 6 ribs Width: 21 mm
10.03	Alternator	1	Aluminium housing, copper windings	Generates electricity for the control unit, ignition, fuel pump and battery charging.	Output: up to 90 A Drive ratio: ≈ 2.7
10.04	Water pump	1	Aluminium housing, impeller	Circulates coolant through the block, heads and radiator.	Flow: ≈ 250 l/min at 7,000 rpm Drive ratio: ≈ 1.6
10.05	Belt tensioner	1	Aluminium arm, coil spring, bearing	Keeps the belt at constant tension.	Belt tension: ≈ 300 N
10.06	Idler pulley	1	Steel, sealed bearing	Guides the belt and increases the wrap on neighbouring pulleys.	Diameter: 64 mm
10.07	Crankshaft position sensor	1	Inductive sensor	The engine’s master sensor: reports crankshaft position and speed.	Trigger wheel: 60 – 2 teeth Type: inductive

ASSEMBLY 11 · 6 items

Dry-sump lubrication

Oil is stored in a separate tank instead of the sump. A multi-stage pump scavenges it from the crankcase and feeds it under pressure to the bearings, so lubrication stays reliable even in long corners at 2–3 g.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
11.01	Dry-sump pan	1	Aluminium alloy, casting	Closes the crankcase from below and gathers the draining oil at the scavenge points.	Height: ≈ 60 mm Scavenge points: 3
11.02	Dry-sump pump	1	Aluminium sections, steel rotors	Scavenges oil from the crankcase to the tank and feeds it from the tank into the engine under pressure.	Stages: 1 pressure + 3 scavenge Oil pressure: 5–6 bar
11.03	Oil pump drive	1 set	Toothed belt with aramid cord, aluminium pulleys	Drives the oil pump from the crankshaft.	Ratio: ≈ 0.6 Belt profile: HTD, 8 mm pitch
11.04	Scavenge lines	1 set	Steel-braided hoses, AN fittings	Connect the sump scavenge points to the pump stages.	Scavenge: AN-12 Pressure: AN-10
11.05	Oil filter	1	Steel canister, synthetic element	Cleans wear debris out of the oil before it reaches the bearings.	Filtration: ≈ 20 µm Bypass valve: ≈ 1 bar

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
11.06	Oil pressure sensor	1	Piezoresistive sensor	Continuously measures oil pressure in the main gallery.	Range: 0–10 bar

ASSEMBLY 12 · 5 items

Flywheel and clutch

Pass torque from the crankshaft to the gearbox and let the driver disconnect them smoothly when pulling away and changing gear.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
12.01	Flywheel	1	Steel, lightened	Carries the clutch pack and the starter ring gear and smooths out rotation.	Diameter: 184 mm Weight: ≈ 3 kg
12.02	Driven plates	3	Sintered-metal facings on steel	Transfer torque from the flywheel to the gearbox input shaft.	Diameter: 140 mm Friction faces: 6
12.03	Intermediate plates	2	Steel	Separate the driven plates and turn with the flywheel.	Thickness: ≈ 3 mm
12.04	Clutch cover	1	Aluminium alloy, steel diaphragm spring	Clamps the plate pack and releases it when the pedal is pressed.	Clamp load: ≈ 6 kN
12.05	Flywheel and cover bolts	14 (8 + 6)	High-strength steel	Fasten the flywheel to the crankshaft flange and the cover to the flywheel.	Flywheel bolts: 8 × M10 Cover bolts: 6 × M8

ASSEMBLY 13 · 3 items

Electronics and wiring

The control unit reads the sensors and calculates injection and ignition timing for every cylinder. The wiring loom connects it to the coils, injectors and sensors.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
13.01	Engine control unit	1	Finned aluminium case, multi-layer board	Controls fuel injection, ignition and the rev limit from the sensor signals.	Injection channels: 8 Ignition channels: 8 Data logging: up to 1,000 times a second
13.02	Wiring loom	1 set	Wires in heat-resistant sleeving, sealed connectors	Carries power and signals between the control unit, coils, injectors and sensors.	Connectors: sealed, latching Temperature rating: up to 150 °C
13.03	Coolant temperature sensor	1	Thermistor in a brass body	Reports engine temperature to the control unit.	Range: –40...150 °C Type: NTC thermistor

ASSEMBLY 14 · 3 items

Oil tank and radiator

In the car these units are mounted away from the engine; on the stand they are shown alongside it. The tank stores the dry-sump oil supply, the radiator passes coolant heat to the oncoming air.

ITEM	PART	QTY	MATERIAL	PURPOSE	SPECIFICATIONS
14.01	Oil tank	1	Welded aluminium alloy	Stores the oil for the dry-sump system and removes air from it.	Capacity: ≈ 10 l Breather: in the top cap
14.02	Oil tank lines	1 set	Steel-braided hoses, AN fittings	Feed oil from the tank to the pressure stage of the pump and return scavenged oil to the tank.	Feed: AN-12 Return: AN-10
14.03	Radiator	1	Aluminium: flat tubes and corrugated fins	Passes coolant heat to the oncoming air.	Working pressure: ≈ 1.5 bar Coolant temperature: 85–100 °C